

CRYPTO STRATEGY LAB (CryptoBot)

Báo Cáo Thiết Kế Kiến Trúc Phần Mềm (Software Architecture)

Nền tảng Research, Auto-Discovery & Đánh giá Trading Strategy Tự động

Trường Đại học Khoa học Tự nhiên - ĐHQG-HCM
Bộ môn Kỹ thuật Phần mềm

Ngày 2 tháng 9 năm 2026

1. Miền Nghiệp Vụ & Bối Cảnh (Business Domain)

[Binance USDT-M Futures & Algo-Trading]

- **Perpetual Futures Market (USDT-M):** Leverage 2 chiều Long/Short 24/7; Order execution theo BBO realtime.
- **Đặc tính tài chính:** Funding Rate (8h), Fee Maker/Taker, Slippage & Rủi ro Liquidation.

[Đối Tượng Sử Dụng & Nhu Cầu Cốt Lõi]

- **Quant Researcher / Trader:** Ingest candle low-latency ($< 200\text{ms}$) đa timeframe (1m, 5m, 1h, 1d); Backtest chính xác (Sharpe, Drawdown, Win Rate).
- **Autonomous AI Agents:** Crawl multi-source news, scoring sentiment; Generate strategy code & scan hyperparameter space.

2. Bài Toán Thực Tế & Thách Thức (Problem Statement)

[Market Data & Order Execution]

- **Realtime Data Jitter & Packet Drop:** WSS rớt kết nối làm mất nền → Auto-detect candle gap & Gap Backfill tự động.
- **Lookahead Bias & Parity Mismatch:** Giả lập thiếu slippage/phí → Deterministic Replay Engine mô phỏng sát thực tế.

[Scalability & Architecture]

- **Combinatorial Explosion:** Quét hàng ngàn tham số gây block UI nếu chạy sync → Async Outbox Job Queue & Worker Pool.
- **Tightly Coupled & Vendor Lock-in:** Code gắn chặt vào 1 sàn → Strategy Plugin Architecture & AST Sandbox.

3. Bối Cảnh & Hệ Thống Quality Attributes (ASRs Taxonomy)

Thuộc tính (QA)	Trọng tâm Thiết kế	Tactic / Kỹ thuật Kiến trúc Áp dụng
Modifiability	Thêm mới Strategy, Search Algorithm, Market Provider không sửa Core	Strategy Plugin Architecture, Open-Closed Principle, Dynamic Registry
Scalability	Xử lý workload > 100,000 backtests & stream candle realtime	Scale-out Python Worker pool, Postgres Leased Job Queue, In-memory Broadcaster
Realtime / Perf	Latency cập nhật candle < 200ms, high throughput Backtest	Go Edge Gateway, WebSocket Streaming, Deterministic Replay Engine
Reliability	Fault isolation (lỗi crawl/search không crash API); auto-reconnect sàn	Failure Isolation, Transactional Outbox, Lease Takeover
Observability	Monitor Worker health, tiến độ Search Loop & telemetry metrics	Structured Logging, State Machine tracking, Run Metrics
Reproducibility	Kết quả Backtest & Leaderboard đảm bảo tính Deterministic & Immutable	Immutable Dataset snapshots, Run Config Hash, Seed Lock

4. Kịch Bản ASR Chi Tiết (Modifiability & Scalability)

[ASR-1: Modifiability (Thêm Strategy mới)]

- **Source:** Quant Researcher / AI Agent.
- **Stimulus:** Thêm strategy class mới (MACDStrategy).
- **Artifact:** Strategy Subsystem & UI Registry.
- **Response:** Auto-discovery, load metadata lên UI không cần compile lại Go Gateway hay sửa Core.
- **Measure:** Thao tác trên 1 file Python độc lập, 0 downtime.

[ASR-2: Scalability (High Backtest Workload)]

- **Source:** User trigger Auto Search Loop.
- **Stimulus:** 10,000 backtest jobs vào Job Queue đồng thời.
- **Artifact:** Job Queue & Python Worker Pool.
- **Response:** Phân phối jobs qua Lease Heartbeat, worker scale-out, zero OOM / memory leak.
- **Measure:** Completion rate 100%, 0 dropped job, stable CPU.

5. Kịch Bản ASR Chi Tiết (Realtime & Reliability)

[ASR-3: Realtime & Data Parity]

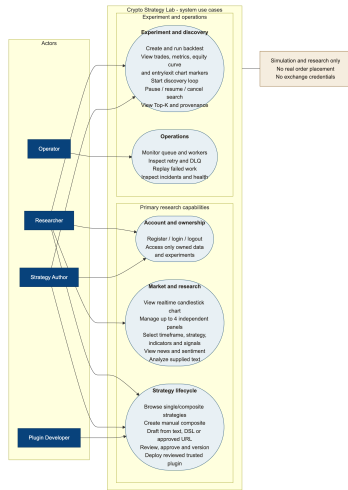
- **Source:** Binance Exchange (WSS drop / network jitter).
- **Stimulus:** Mất kết nối mạng trong 30 giây.
- **Artifact:** Go Market Gateway & Postgres Candle Storage.
- **Response:** Auto-reconnect, trigger REST Backfill bù missing candles, deduplicate bằng Open Time.
- **Measure:** Continuous candle stream, 0 duplicate, catch-up latency < 2s.

[ASR-4: Fault Tolerance (Worker Crash)]

- **Source:** Python Worker process bị kill đột ngột (OOM).
- **Stimulus:** Job đang ở trạng thái RUNNING.
- **Artifact:** Job Queue & Outbox Manager.
- **Response:** Lease Heartbeat timeout (30s), Worker khác auto-takeover và retry job.
- **Measure:** Job hoàn thành thành công, 0 stuck / orphaned job.

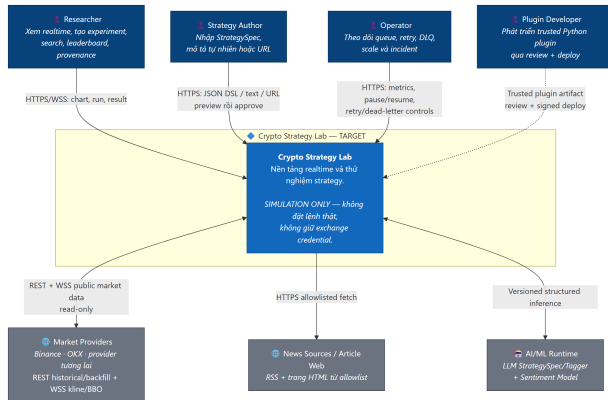
6. Tổng Quan Use Case Hệ Thống

- **Quant Trader:** Xem realtime candlestick chart đa timeframe, authoring strategy, chạy backtest, xem leaderboard.
- **AI Agent:** Crawl financial news, extract text, scoring sentiment, trigger Loop Discovery tìm strategy tối ưu.
- **System Worker:** Ingest candle định kỳ, process async Backtest Job Queue.



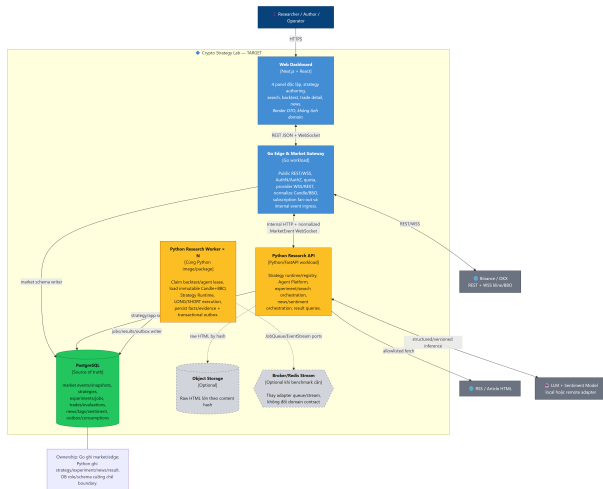
7. C4 Model (Level 1) - System Context Diagram

- **CryptoBot Core Platform:** Platform trung tâm cho market data ingestion, strategy research và portfolio ranking.
- **External Systems:**
 - **Binance Exchange:** Historical REST & WebSocket BBO price stream.
 - **News / RSS Feeds:** Multi-source crypto news feeds.
 - **LLM Providers (OpenAI/Groq):** LLM reasoning, sentiment analysis & strategy self-repair.



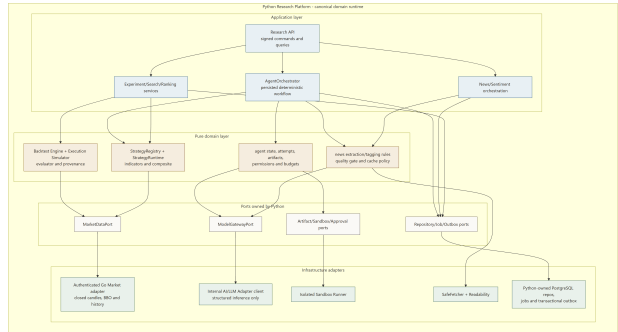
8. C4 Model (Level 2) - Container Diagram

- **Next.js Dashboard:** Render-only UI cho chart, authoring, search, backtest, trade detail & news.
- **Go Edge & Market Gateway:** Public REST/WSS, Auth/Quota, Candle/BBO normalization & realtime fan-out.
- **Python Research API:** Strategy/Agent runtime, experiment/search, news/sentiment orchestration & queries.
- **Python Research Worker × N:** Leased backtest/agent jobs; immutable Candle+BBO; execution; facts/outbox.
- **PostgreSQL:** Source of truth: market, strategies, jobs/results, news/sentiment & outbox.
- **Object Storage / Broker (optional):** Raw HTML for the content.



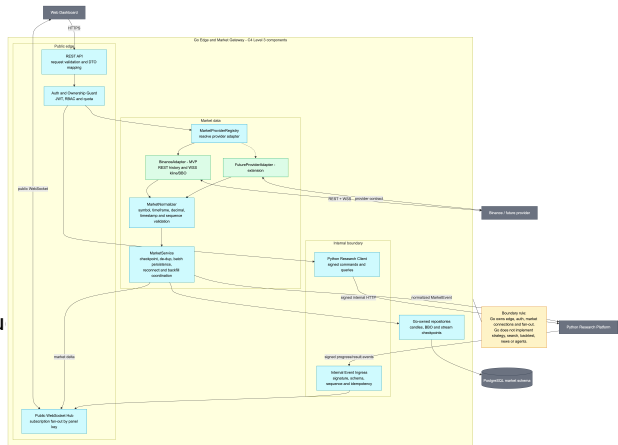
9. C4 Model (Level 3) - Python Research Platform

- **Application services:** Research API, Experiment/Search/Ranking, News/Sentiment & AgentOrchestrator.
- **Domain runtime:** StrategyRegistry + StrategyRuntime; Backtest Engine + Execution Simulator; news rules; agent state/artifacts.
- **Python-owned ports:** MarketDataPort, ModelGatewayPort, Artifact/Sandbox/Approval & Repository/Job/Outbox ports.
- **Infrastructure adapters:** Go Market adapter, internal AI/LLM adapter, Sandbox Runner, SafeFetcher/Readability & PostgreSQL repositories.



10. C4 Model (Level 3) - Go Edge & Market Gateway

- **REST API + Auth Guard:** Request validation/DTO mapping, JWT, RBAC & quota.
- **Public WebSocket Hub:** Subscription fan-out the panel key.
- **MarketProviderRegistry + BinanceAdapter:** Resolve provider, REST history & WSS kline/BBO.
- **MarketNormalizer + MarketService:** Validate symbol/timeframe/decimal/timestamp/sequence checkpoint, de-dup, persistence, reconnect & backfill.
- **Python Research Client + Internal Event Ingress:** Signed commands, queries & progress/result events.



11. Component Boundaries: 6 Module Cốt Lõi (Anti-God Service)

[6 Bounded Contexts (High Cohesion)]

- ➊ **Market Realtime (Go Edge):** WSS candles/BBO, normalization, gap backfill.
- ➋ **Strategy Engine (Python):** Single/composite runtime qua Registry.
- ➌ **Backtest (Python Worker):** Candle+BBO replay, LONG/SHORT execution, trade facts.
- ➍ **Discovery / Search Loop (Python):** Candidates, jobs, validation, lineage, stop conditions.
- ➎ **News Crawler (Python):** RSS/HTML, SSRF guard, quality gate.
- ➏ **News Intelligence (Python + AI)**

[Trách Nhiệm Công Nghệ (Go vs Python)]

Tiêu chí	Go Edge & Market	Python Platform
Ownership	Market Ingestion, Edge API, fan-out	Strategy, Backtest, Discovery, News/Agents/Ranking
Strengths	Goroutines non-blocking, Low RAM	Quantitative math, AST, ML
Protocols	Public REST/WSS, signed events	Internal HTTP, DTOs, Job/Outbox Ports
Isolation	Lỗi network	Crash strategy

12. Đặc Tả Ranh Giới & Phạm Vi 6 Module (Component Specs)

[1. Market Realtime & 2. Strategy Engine]

- **Market Realtime:** Raw Binance WSS → Normalized Candle & BBO; persist vào Postgres, broadcast CandleClosed qua SSE/WS; Invariant: continuous time-series, zero-duplicate.
- **Strategy Engine:** Pluggable qua IStrategy; Output là Signal (BUY/SELL/HOLD); 100% execution parity Live/Backtest.

[3. Backtest & 4. Discovery]

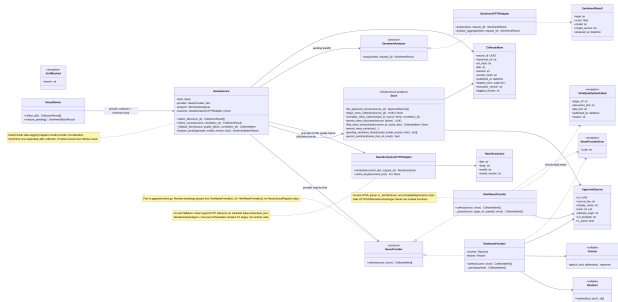
- **Backtest Subsystem:** Replay immutable Candle+BBO với cùng runtime; simulate execution & emit trade facts qua Worker/Outbox.
- **Discovery Subsystem:** Generate candidates, orchestrate Search Loop qua Train/Validation/Sealed Test; enforce lineage & stop conditions.

[5. News Crawler & 6. News Intelligence]

- **News Crawler:** Fetch news từ ApprovedSource (RSS/HTML), SSRF protection, Quality Gate validation

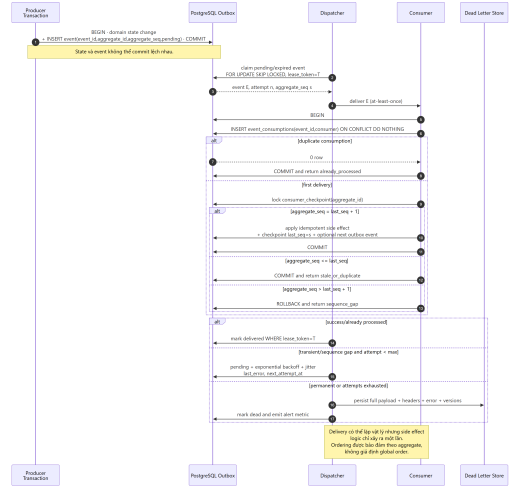
15. UML Class Diagram: Resilient News Crawler Model

- **Contract NewsProvider:** Implemented by RssNewsProvider & HtmlNewsProvider (ApprovedSource).
- **SSRF Guard & Quality Gate Pipeline:** Resolver & Fetcher validate DNS, private IP blocking, safe redirect.
- **LLM Fallback Extraction:** Trigger NewsExtractionHTTPAdapter khi DOM thay đổi.
- **Decoupled Sentiment Scoring:** Async batch scoring, AI errors không block crawl pipeline.



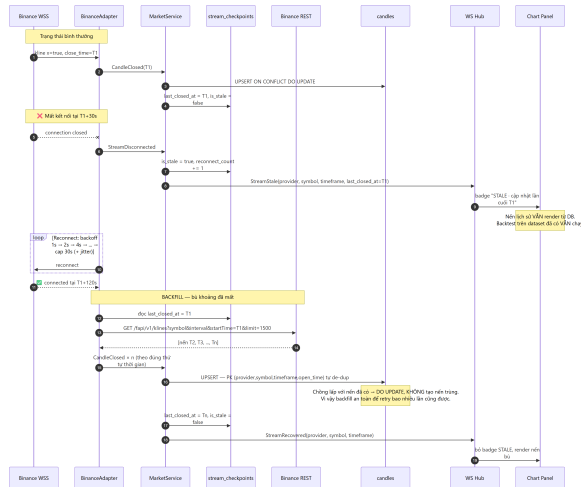
17. Các Design & Architectural Patterns Trọng Yếu

- **CQRS (Go API):** Tách biệt Command (tạo experiment) và Query (đọc candles, leaderboard), optimize latency.
- **Transactional Outbox:** Atomic consistency khi tạo Experiment và dispatch Job, eliminate dual-write risk.
- **Dual-Channel Market Engine with Parity:** Schema & execution parity cho Realtime WSS và Historical Backfill.
- **Plugin Architecture:** Decouple hoàn toàn Core Engine khỏi strategy logic và search algorithms (Open-Closed Principle).



19. Runtime Flow: Realtime Ingestion & Gap Backfill

- Chart Initialization:** Fetch 1,000 historical candles từ Postgres/Binance REST đa timeframe (1m, 5m, 1h, 1d).
- WSS Stream Ingestion:** Ingest ticker & update provisional candle theo realtime ticks.
- Candle Close Event:** Persist candle vào Postgres và broadcast event CandleClosed qua SSE/WS.
- Gap Backfill:** Khi WSS drop, auto-reconnect và trigger REST API backfill bù missing candles.



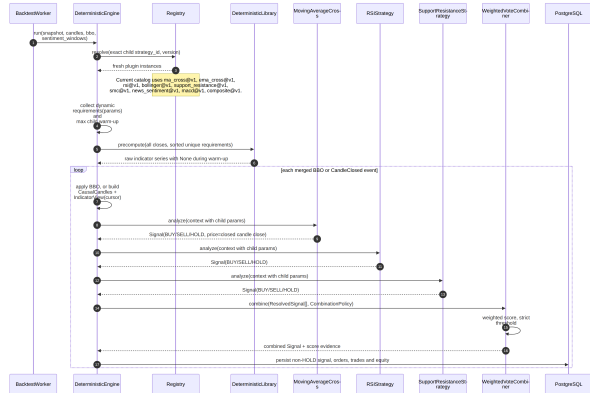
20. Runtime Flow: Strategy Execution & Dynamic Registration

- **Run Strategy Flow:**

- Load Datafeed & Sentiment Window vào StrategyContext.
- IStrategy.evaluate() emit trading signal BUY / SELL / HOLD.
- CompositeStrategy aggregate signals theo majority/weighted policy.

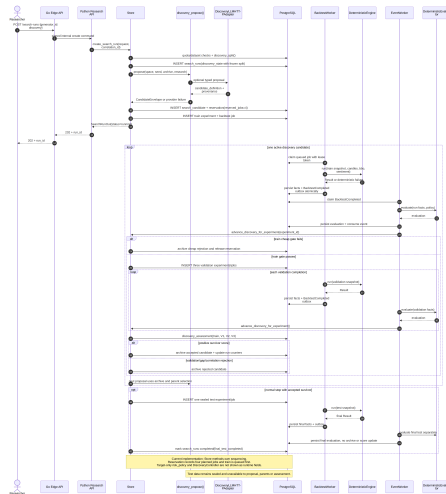
- **Add Strategy Flow:** Handcrafted (drop Python file → auto-load) & AI-Generated (LLM → AST Sandbox).

- **Runtime Parity:** 100% deterministic parity giữa Live Trading và Backtest execution.



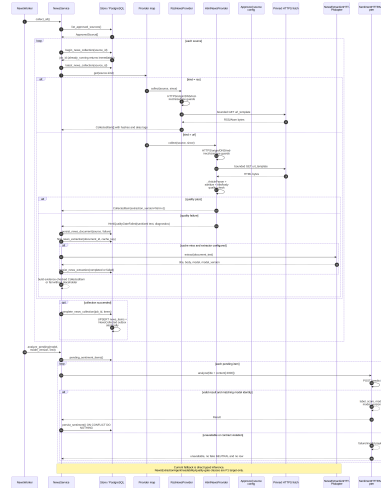
21. Runtime Flow: Async Backtest Pipeline

- 1 **Create Experiment:** Persist metadata và job event vào Outbox trong cùng một ACID transaction.
- 2 **Worker Lease Acquire:** Worker claim job qua Optimistic Lock (FOR UPDATE SKIP LOCKED) & gửi heartbeat.
- 3 **Multi-Split Execution:** Run backtest trên Train → Validation → Sealed Test.
- 4 **Compute Metrics & Leaderboard:** Calculate Sharpe Ratio, Max Drawdown, Profit Factor; persist Trade logs & update Leaderboard.



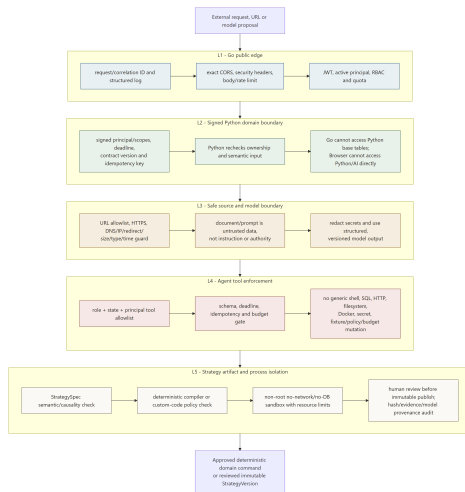
22. Runtime Flow: Resilient News Crawl & LLM Fallback Pipeline

- 1 **SSRF Validation:** Whitelist check qua ApprovedSource, resolve DNS & block internal/private IPs.
- 2 **Standard Ingestion:** Fetch & extract article content qua RSS/HTML Parser (Readability).
- 3 **Quality Gate Check:** Trigger fallback nếu DOM thay đổi hoặc extraction rỗng.
- 4 **LLM Fallback Extraction:** Dispatch raw HTML sang LLM Agent để parse structured content.
- 5 **Async Sentiment Scoring:** Background worker scoring sentiment [-1.0, 1.0] và persist vào PostgreSQL.



23. Security Architecture: Defense-in-Depth

- **Authentication & RBAC: JWT**
authentication, role-based access control, tenant/user experiment isolation.
- **Crawler SSRF Prevention:** Block private IP ranges (127.0.0.1, 10.0.0.0/8, cloud metadata), domain whitelisting qua ApprovedSource.
- **AST Sandbox & Safe Execution:** AST static analysis trên AI-generated code; ban dangerous builtins/modules (eval, subprocess, socket, os.system).
- **Tool Invocation Boundary:**
Schema-validated DTOs, strict input boundaries cho AI Agents.



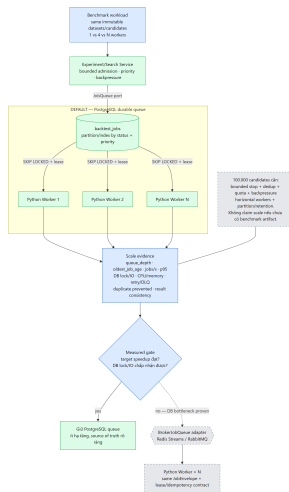
24. Scalability Architecture & Benchmark

• Stateless Horizontal Scale-Out:

- Spin up N Python Worker instances độc lập dựa trên CPU load / queue backlog.
- Distribute jobs qua PostgreSQL B-Tree indexed Leased Job Queue.

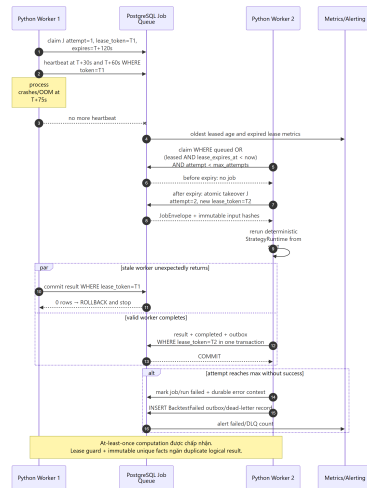
• Load Testing & Benchmarking (k6 / Locust):

- Simulate 10,000 concurrent users dispatch Backtest jobs & query candle data.
- Queue throughput đạt $> 1,500$ backtests/phút trên 4 workers.
- Candle & Leaderboard API latency duy trì $\leq 120\text{ms}$ (p95).



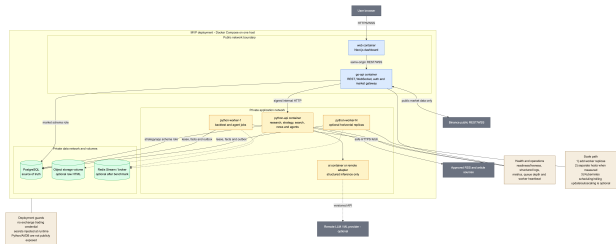
25. Fault Tolerance & Self-Healing Architecture

- **Worker Lease Takeover & Heartbeat:** Heartbeat 10s/lần. Nếu worker crash, lease timeout sau 30s, worker khác auto-takeover job.
- **Idempotency & Retry:** Unique idempotency_key constraint, eliminate duplicate execution risk.
- **Failure Isolation & Reconnect:** Plugin failure isolation (lỗi 1 strategy không crash Worker); WSS auto-reconnect & backfill.
- **Chaos Engineering Simulation:** Kill worker process / inject network drops để verify self-healing behavior.



26. Deployment Topology & MLOps Infrastructure

- **Full-Stack Containerization:** Isolated container services: Next.js Web Dashboard, Go Edge Gateway, Python Research API, Python Research Worker $\times$ N, Internal AI Inference Adapter, PostgreSQL.
- **Health Checks & Liveness Probes:** /healthz và /readyz endpoints cho container auto-restart & traffic routing.
- **MLOps & Configuration:** Versioned system prompts và LLM API keys tách biệt qua environment variables (.env).
- **Kubernetes-Ready Architecture:** Hỗ trợ Deployment Replicas, Rolling Updates, HPA khi scale production.

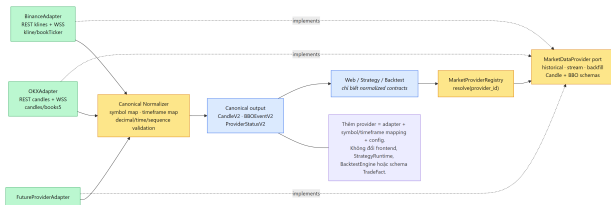


27. Architectural Tradeoffs Matrix

Lựa Chọn Thiết Kế	Phương Án Được Chọn	Phương Án Thay Thế	Lý Do & Đánh Đổi Kiến Trúc (Rationale)
System Architecture	Modular Monolith	Microservices	Giảm overhead vận hành mạng/distributed latency; enforce module boundaries qua contracts.
Candle & Job Storage	PostgreSQL + B-Tree	ClickHouse / InfluxDB	Giữ tính ACID cho Experiments & Outbox; B-Tree indexing đủ đáp ứng hàng triệu candles.
Job Queue & Event Ingress	PostgreSQL Outbox & Leases	Kafka / RabbitMQ / Redis	Loại bỏ dual-write, ACID transactional guarantees, zero additional infrastructure dependencies.
AI Code Execution	AST Analyzer + Sandbox	Docker-in-Docker	Low latency initialization (< 10ms), static syntax safety check, minimal container resource overhead.
Sentiment Analysis Pipeline	Hybrid: Rule + LLM Batch	Pure LLM Realtime	Optimize token cost & API rate limits; non-blocking crawl pipeline, deep LLM scoring theo batch.

28. Replaceability & Extensibility Architecture

- **Pluggable Market Data Providers:**
MarketProviderAdapter interface cho phép switch từ Binance sang OKX, Bybit, Coinbase mà **không thay đổi** Core Frontend hay Strategy Engine.
- **Extensible Search Algorithms:** Sẵn sàng plug-in Reinforcement Learning (PPO) hoặc Custom Optimizers qua interface ISearchAlgorithm.
- **Extensible News Ingestion:** Dễ dàng add new sources (CryptoPanic, CoinDesk API) qua config ApprovedSource.



29. Architectural Summary & Project Conclusion

[Architectural Highlights & Key Takeaways]

- ✓ **ASR-Driven Architecture:** Bám sát 4 Architectural Drivers & Quality Attributes taxonomy.
- ✓ **Clean Boundaries & Anti-God Service:** Phân rã 6 bounded contexts, zero tight-coupling.
- ✓ **Multi-Agent & Automation:** Self-repair loop (AST + Sandbox) & anti-overfitting data partitioning (Train/Val/Test).
- ✓ **High Scalability & Self-Healing:** Scale-out 100,000 backtests, Lease Takeover & Idempotent execution.

[Architecture Artifacts & Verification]

- **39 Blueprint Diagrams & Specs:** C4 Model (L1-L3), UML Class, State Machines, Sequence Flows, Deployment Topology.
- **5 Interactive Working Dashboards:**
 - ① Realtime Market Chart & Indicators
 - ② Strategy Authoring & Plugin Registry
 - ③ Async Backtest & Trade Visualization
 - ④ Search Loop & Top-K Leaderboard
 - ⑤ News Crawler & Sentiment Intelligence
- **Automated Test Scenarios & Benchmark Suite.**